- The magnetic field around a magnet is represented by imaginary lines called magnetic line of force.
- The magnetic line of force emerges from north pole to south pole and it continues through the body of magnet to form a closed loop.
- Two magnetic lines of force will not intersect each other.
- If magnetic lines of force are rows together, the field is strong. If they are spaced out the magnetic field is weak.

b) Magnetic Flux

- The amount of magnetic field produced by a magnetic source is called magnetic flux.
- It is denoted by Greek Letter ϕ and its unit is weber.

c) Magnetic Flux Density

- The magnetic flux density is the flux per unit area at right angles to the flux.

Magnetic flux density, $B = \phi/A$ wb/m^2

d) Permeability

- Permeability of a material means, the conductivity for magnetic flux. The greater the permeability of material, the greater is its conductivity of magnetic flux and vice-versa. Air or Vacuum is the poorest conductor of magnetic flux. The absolute (actual) permeability μ_0 (Greek Letter 'mu') of air is $4\pi \times 10^{-7}$ Henry/metre. The absolute (actual) permeability of magnetic material(μ) is much greater than μ_0.

The ratio between permeability of material and permeability of air (μ_0) is called relative permeability(μ_r)

$$\mu_r = \mu/\mu_0$$

The relative permeability for air is 1 ($\mu_r = \mu_0/\mu_0$)

The value of μ_r for all non- magnetic material is also 1.

The relative permeability of magnetic materials is very high. For example, soft iron (i.e pure iron) has a relative permeability of 8000, whereas its value for perm alloy (22% Iron, +78% nickel) is as high as 50,000.

e) Magneto Motive Force (MMF)

It is a magnetic pressure which tends to set up magnetic flux in a Magnetic circuit.

The work done in moving a unit magnetic pole once round the magnetic circuit is called MMF. It is equal to the product of current and number of turns of the coil.

MMF = Number of turns × current.
Its unit is Ampere-turns

f) Reluctance

The opposition that the magnetic circuit offers to magnetic flux is called reluctance. Magnetic materials (eg iron, steel) have low reluctance, on the other hand non-magnetic materials have a high reluctance.

Reluctance $S = l/ \mu_0 \mu_r A$

g) Magnetic Neutral Axis (MNA)

The imaginary line which is perpendicular to the magnetic axis and

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